

Assignment 2 Answer Key

1. (a) Line Voltages: $207.8\angle 60^\circ \text{ V}$, $207.8\angle -60^\circ \text{ V}$, $207.8\angle -180^\circ \text{ V}$
(b) Line Currents: $3.75\angle -8.66^\circ \text{ A}$, $3.75\angle -128.66^\circ \text{ A}$, $3.75\angle 111.34^\circ \text{ A}$
2. Phase currents: $6\angle -60^\circ \text{ A}$, $6\angle -180^\circ \text{ A}$, $6\angle 60^\circ \text{ A}$
Line currents: : $10.392\angle -90^\circ \text{ A}$, $10.392\angle -210^\circ \text{ A}$, $10.392\angle 30^\circ \text{ A}$,
3. $I_{AB} = 5.548\angle 65^\circ \text{ A}$, $V_{AB} = 120\angle 98.69^\circ \text{ V}$
4. Line Currents: $7.21\angle -66.34^\circ \text{ A}$, $7.21\angle 173.66^\circ \text{ A}$, $7.21\angle 53.66^\circ \text{ A}$
5. 46.31 A
6. Line Currents: $64\angle 80.1^\circ \text{ A}$, $38.1\angle -60^\circ \text{ A}$, $42.5\angle 225^\circ \text{ A}$
Real Power: 4.84 kW
7. $Q = 50$, $\omega_0 = 100 \text{ krad/s}$, $\omega_1 = 99 \text{ krad/s}$, $\omega_2 = 101 \text{ krad/s}$,
 $B.W. = 2 \text{ krad/s}$
8. 435.9 rad/s
9. 7.07 A
10. $R = 50 \Omega$, $L = 10.61 \text{ mH}$, $C = 1.061 \mu\text{F}$